Submersible Pump



Category: **Plumbing & Pipe Fitting** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Submersible pump

What it is used for

A submersible pump operates while immersed in water or another manufacturer-approved liquid to move it through a discharge pipe. Correct installation, electrical protection, and dry-running prevention help maintain safe and reliable operation.

Main parts

- Pump body and sealed motor
- Water inlet and strainer
- Discharge outlet and pipe connection
- Power cable and plug or terminal connection
- Float switch or level sensor if fitted
- Check valve and lifting rope or chain where specified

Before you start

1. Confirm the liquid, temperature, depth, and pump capacity are within the manufacturer's limits.
2. Inspect the pump body, cable, plug, seals, and discharge connection for damage.
3. Check the supply voltage, earthing, and residual-current protection as required.
4. Verify the intake strainer and discharge pipe are clear and correctly connected.
5. Ensure the pump is fully submerged to the required level before starting, unless designed otherwise.
6. Secure a suitable lifting rope or chain to the designated lifting point; never use the cable.

How to use

1. Place the pump on a stable base clear of mud, debris, and loose sediment.
2. Connect the discharge pipe and secure all required clamps or fittings.
3. Lower the pump using the designated lifting rope or chain without dropping it.
4. Confirm the pump's minimum immersion and cooling requirements are satisfied.
5. Set the float switch or level sensor so it can move freely, if fitted.
6. Switch on the supply using the approved control and observe the initial discharge.
7. Check for abnormal noise, vibration, leaks, overheating, or poor flow.
8. Stop the pump if the water level falls below its permitted operating level.
9. Switch off and isolate the electrical supply before removing or repositioning the pump.
10. After use, clean the strainer and exterior, inspect the cable, and store as instructed.

⚠ Safety

- Never lift, lower, or carry the pump by its electrical cable.
- Do not touch the pump, plug, or water with wet hands while connected to power.
- Use appropriate earthing and residual-current protection installed by a qualified person.
- Never enter a tank, pit, or confined space to retrieve a pump without an approved safe-work procedure.
- Do not run the pump dry unless the model is specifically designed for dry operation.
- Isolate and verify the electrical supply before maintenance or clearing a blockage.

Care & storage

- Clean the intake strainer and pump exterior after use as required.
- Inspect the power cable, seals, and discharge fittings for wear or damage.
- Follow the manufacturer's schedule for servicing bearings, seals, and other serviceable parts.
- Store the pump in a clean, dry, frost-protected place when required by its design.

Common mistakes to avoid

- Running the pump dry can overheat and damage the motor or seals.
- Lifting the pump by its cable can damage insulation and create electric-shock hazards.
- Placing the intake in mud or debris can block flow and overload the pump.
- Using an unsuitable extension lead or missing electrical protection increases shock risk.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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